

The Red Queen's Prison

Frame-Lock and the Architecture of Organizational Collapse in Evolutionary Warfare

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Part 3 of a trilogy on substrate-independent life. Companion to “Turtles, or Crystals” (Schulz, 2026) [1] and “Turtles, All the Way Up” (Schulz, 2026) [2]. Papers 1–2 asked how a substrate-independent living organization could work; this paper asks why such an organization fails when it locks onto a single coordination substrate, and how it can escape. The three papers are meant to reference and correct one another.

This is a speculative cross-domain synthesis. It draws on military history, information theory, organizational science, and the evolutionary notion of the Red Queen; the individual results it leans on are established, but the bridges between them are conjecture. We present frame-lock as an organizing lens, not a law. We mark every seam, pull out the one claim we are most willing to be wrong about (§8), and state the experiment that tests it. A first, topology-only run of that experiment is now reported (§8.2, Appendix A); its figures are results about the model, not about the world, and the broader mission-command thesis remains untested.

Abstract

Some systems optimize so completely for one mode of coordination that they cannot change modes even when changing is the only way to survive. We call this state *frame-lock* and offer it as a lens linking cases that are not usually read together: a submarine fleet whose every coordinated success ran through one ultimately transparent cipher, and a centralized command structure in which a single corrupted node becomes a cascade. We propose that frame-lock’s most dangerous form is what we name the *False Run* — a system’s confident but mistaken belief that its substrate is secure when it is in fact compromised. Under that belief a system does not merely fail to defend itself; it leans *harder* on the compromised channel, trusts corrupted nodes *more*, and walks into prepared traps while believing it is winning.

We sketch five layers by which frame-lock deepens — substrate colonization, hidden goals, cascading corruption, signal flooding, and the trusted saboteur — and argue from worked examples that the layers compound. We are explicit throughout that these are arguments from analogy and documented history, not new demonstrations. We then state the one claim the paper stakes its credibility on, as a falsifiable prediction with its experiment: that decision architectures specifying *intent* rather than detailed orders (the tradition militaries call mission command) are measurably more robust to corruption and flooding than central-

ized command. We give the simulation that tests it and report a first result: in a topology-only experiment the predicted crossover appears — centralized coordination is more accurate when the channel is clean, but a single corrupted central node collapses it while a distributed arm degrades gracefully — for the modelled mechanism only; the broader thesis remains open. Released under CC BY 4.0 as a lens to argue with, and to break.

PART 1 · THE PRISON

1. Introduction: why capable systems get trapped

In Lewis Carroll's *Through the Looking-Glass*, the Red Queen tells Alice it takes all the running one can do to stay in the same place. Evolutionary biology borrowed the image for the Red Queen hypothesis [3]: lineages locked in coevolutionary arms races must keep adapting merely to hold position. This paper is about a failure mode one level deeper. Not the exhaustion of running in place, but the trap of having optimized so hard for one way of running that no other gait remains — even when the ground has changed and the old gait is now fatal. We call that trap the Red Queen's Prison, and the structural condition behind it frame-lock.

Our anchor case is documented and unambiguous. In 1940–1943, German submarine “wolfpacks” coordinated their attacks on Atlantic convoys through radio traffic enciphered by the Enigma machine. The coordination was the innovation, and for a time it was devastatingly effective. But the coordination depended entirely on that one channel — and once the Allies were reading it, the channel that produced the success produced the slaughter. Critically, the German command did not know its cipher was broken; believing it secure, it kept using it. The fleet was imprisoned by the very optimization that had made it formidable, and could not see the walls. That perceptual gap — not the loss itself — is the heart of this paper.

We use a second, contemporary case for vividness and flag it carefully as contested: a highly centralized command structure under stress, where a single corrupted order or a single trusted-but-degraded node can propagate through the whole formation. We develop the anchor case in §6 and the contested one in §7, after laying out the mechanism. Both are illustrations of one idea: a system can be brilliant inside its frame and helpless the moment the contest moves outside it.

2. The central conjecture: the False Rung

Before any of the mechanisms below can do its work, one condition usually has to hold: the system must not know that its channel is compromised. We call this the False Rung — the unexamined belief that the substrate one is standing on will bear weight, when it will not.

The conjecture is simple to state. Awareness of compromise is itself a defense, because it unlocks options that are otherwise invisible:

- If a system knows its substrate is readable, it can change substrates, or assume it is read and behave accordingly.
- If it knows a node is corrupted, it can isolate it.
- If it knows it is being flooded, it can treat the flood as adversarial rather than as ordinary load.

Deny the system that knowledge and the failures invert into something worse than passive vulnerability. Believing the channel secure, the system uses it *more* confidently; believing the central node sound, it trusts that node *more*, not less. Confidence becomes the mechanism of harm.

2.1 The historical anchor: Ultra and the Enigma secret

The clearest historical case is the Allied reading of Enigma. The decisive feature was not only that Bletchley Park could read German naval traffic, but that the Allies worked hard to keep Germany from learning it. Acting too freely on decrypts would have revealed that the cipher was broken; so the intelligence had to be rationed, and its fruits attributed to plausible alternative sources — reconnaissance flights, agents, luck. Historians of the campaign record that this discipline sometimes meant *not* acting on foreknowledge of an attack, because acting would have cost the larger secret [6]. (We state this as the well-documented shape of the Ultra problem; we do not put specific words in any individual's mouth, and readers should consult the official history for detail.)

The German side furnishes the False Rung in its purest form. Confronted with mounting, suspiciously well-targeted losses, German command considered that Enigma might be compromised — and concluded instead that the machine was sound and that a traitor or conventional intelligence was responsible. The belief in the cipher held. So the channel kept being used with confidence, and the foreknowledge on the other side kept being exploited. The substrate was transparent; the owner did not know; and not knowing, the owner leaned on it harder.

3. The architecture: five layers, one false belief

We describe frame-lock as five layers that can stack, each made more dangerous by the False Rung. We present the layers as an organizing scheme; the claim that they *compound* — that vulnerability at one layer raises vulnerability at the next — is a hypothesis, sharpened into a testable form in §8.

- **Layer 1 — Substrate colonization.** One coordination channel comes to carry essentially one function, so well that alternatives never develop. The system cannot switch without surrendering the advantage that defines it.
- **Layer 2 — Hidden goals.** Subordinate parts have aims that diverge from the whole. Centralized structures can monitor outputs but cannot read intent; distributed structures hide divergence but make it self-punishing.

- **Layer 3 — Cascading corruption.** A node that is degraded but still functioning is, we argue, more dangerous than one that is destroyed: the system keeps trusting it, and routes decisions through it.
- **Layer 4 — Signal flooding.** Overwhelming the channel with plausible traffic can push a system into a reactive, defensive posture in which it can no longer tell signal from noise.
- **Layer 5 — The trusted saboteur.** An insider with legitimate access and a trusted position can corrupt from within, and — this is the unsettling part — the system’s own trust architecture shields the saboteur from detection.

In each case the False Rung is the enabling condition: the system does not defend against a layer it does not believe it is exposed to. The moment it accepts that compromise is possible, the rung is no longer false, and the options in Part 3 open up.

4. Layer 1: substrate colonization

When a system optimizes for a single mode of coordination, that channel can become so specialized to the mode that it has no spare capacity for anything else — and, just as important, no fallback if the channel fails. The wolfpack is the clean illustration. Submerged boats cannot see one another and dare not signal acoustically without revealing position; radio was effectively the *only* medium through which a dispersed pack could be concentrated onto a convoy. The doctrine that made the boats deadly therefore committed them to that one channel. There was no second coordination substrate to fall back on, so when the channel was compromised there was nowhere to go: radio silence preserved security but dissolved the coordination, and every alternative (cipher changes, acoustic methods, new boat designs) was either overtaken or years away. The substrate had been colonized by the doctrine, and the doctrine had no exit.

Stated as a general principle, and as a lens rather than a law: a system that retains more than one independent way to coordinate — and that treats “there may be a better way, and this one may be compromised” as a standing assumption — is structurally harder to lock. This is as true of an engineered network with redundant, dissimilar links as of any social one. Redundancy of *kind*, not merely of capacity, is what resists Layer 1: two copies of the same channel share the same single point of failure; two unlike channels do not.

5. Layer 3: cascading corruption

We treat Layer 3 in detail because it carries the paper’s sharpest mechanism and its most counter-intuitive claim: that corruption can be worse than destruction.

5.1 Why a degraded node can be worse than a dead one

A destroyed node is, informationally, clean: the system registers the loss and routes around it. A node that is degraded but still reporting is dirty: it keeps presenting itself as authoritative, the system keeps trusting it, and the system may even divert resources to protect it. The asymmetry is what makes a mobility

kill so valuable in armored warfare — disabling a vehicle while leaving it in the formation can impose more cost than destroying it outright, because the formation slows, bunches, and exposes itself to defend the casualty. We offer this as a mechanism with a clear logic, not as a measured result; the experiment in §8 tests how centralized and distributed architectures handle the corruption case directly, while a clean destruction-versus-corruption comparison remains future work (§9).

5.2 The cascade, stage by stage

1. **Corruption.** A node is degraded but still broadcasting. Under the False Rung, the system does not register the change.
2. **Trust amplification.** Because the node is central, the system weights its (now corrupted) output heavily.
3. **Propagation.** Downstream nodes act on the corrupted output because it carries the authority of the centre.
4. **Self-defense.** Noticing strain, the system spends effort preserving the very node that is the source of strain.
5. **Collapse.** By the time corruption is recognized, the cascade is already advanced.

The schematic example used throughout the project is a mobility-killed vehicle in a formation: a disabled but living unit reports a degraded position, the formation slows to keep cohesion, the whole group's tempo collapses to the speed of its slowest member, and effort spent shielding the casualty exposes the rest. The unit that was never destroyed becomes the instrument by which the formation is. We present this as an illustrative scenario, not a claim about any specific engagement.

PART 2 · WHY THE PRISON HOLDS

6. Case study: the U-boat wolfpack, locked and lost

6.1 Dominance, 1940–1942

Wolfpack tactics — multiple boats vectored onto a convoy by central direction and attacking in concert — were a genuine operational innovation, and for a period they were devastatingly effective. The coordination depended on radio traffic to and from headquarters, enciphered by Enigma. 1942 was the worst year of Allied shipping losses of the war.

6.2 The False Rung in action, 1943

Once Allied codebreaking matured, that traffic was being read. Convoy routing could be adjusted and escort groups positioned where the boats were directed to go. The reversal was stark: in May 1943 (“Black May”), German losses spiked to roughly forty boats in the month, and the offensive was effectively withdrawn. The decisive point for our argument is the German command's belief: persuaded that Enigma was secure (see §2.1), it attributed the losses to other causes and

kept using the channel. The substrate was transparent; the owner did not know; the confidence itself deepened the damage.

6.3 Why escape was foreclosed

Every available alternative cut against the source of the advantage. Radio silence preserved security at the cost of the coordination that made wolfpacks work. Cipher changes were overtaken. Acoustic coordination would have advertised position. New boat designs lay years away. The fleet was, in our terms, locked at Layer 1 (one coordination substrate), exposed at Layer 3 (each attempted fix compromised in turn), and standing on the False Rung throughout. It is the cleanest documented instance we know of a coordination channel that was simultaneously the whole of an advantage and the whole of a vulnerability.

7. Case study (contested): centralized command in contemporary war

We include a contemporary case for vividness while flagging, more strongly than for the anchor case, that it rests on contested, incomplete, open-source reporting from an ongoing conflict. It should be read as a hypothesis the lens generates, not as an established account, and not as a partisan claim.

The lens predicts that a highly centralized force is fast in stable conditions and fragile in a particular way under stress: a single corrupted order, a single turned or incompetent node trusted because of its position, or spoofed reporting accepted because it flows through the proper channel, can propagate. It further predicts that such a system, under the False Rung, will respond to early failure by centralizing *further* — tightening control over a channel it does not realize is compromised — and will tend to discount the possibility of internal corruption. By contrast, a force that pushes decision authority downward, keeps several independent communication paths, and judges subordinates by whether they achieved the intent rather than whether they executed a specific order, should degrade more gracefully.

Reporting from the 2022–2024 war in Ukraine has been read by many analysts as consistent with this contrast — a more centralized force struggling against more distributed, improvisational opponents. We note that reading and its fit to the lens, and we explicitly decline to assert the internal facts (corruption levels, loyalties, motives) that such reporting alleges but cannot, from the outside, establish. The case is here to illustrate the prediction, and to be discarded if the prediction proves to misdescribe events.

PART 3 · THE ESCAPE, AND THE CLAIM WE WILL DEFEND

8. The testable core: intent-based command

A synthesis earns engagement only where it can be refuted. Most of this paper is a lens; this section is the part we are willing to be wrong about, stated as prediction, experiment, and falsification — in the manner of the companion papers.

8.1 The idea

The escape from frame-lock, we argue, is not stronger defenses at each layer but a decision architecture that does not require any single piece of information to be correct. Centralized command specifies the *how* — position, time, route, force — and therefore fails if any of those is corrupted. Intent-based command (historically, the tradition of *Auftragstaktik* or mission command [4], and the spirit of Boyd’s work on command and control [5]) specifies the *what* — the objective — and delegates the how. A unit that holds the intent can reject an order that contradicts it, adapt when a route is blocked, and continue when the centre falls silent. Crucially, such a system behaves correctly *whether or not* it knows it is compromised: it does not depend on the rung being sound. That is why we tie the escape to the False Rung rather than to better encryption.

Claim (the flag). Decision architectures that specify intent and delegate execution are measurably more robust to cascading corruption (Layer 3) and signal flooding (Layer 4) than architectures that specify detailed orders, for the same task and resources.

Experiment. In an agent-based simulation (appendix), instantiate two populations identical except in decision architecture — one command-following, one intent-based — and subject both to the same scripted attacks (jamming, spoofing, partial-compromise/mobility-kill, structural decapitation, signal flood, and combinations). Measure a coherence/effectiveness metric during attack and recovery after it, across many random seeds, and report effect sizes with variation.

Falsification. If the intent-based architecture shows no robustness advantage across realistic attacks and parameterizations — or if a centralized architecture with cheap monitoring matches it — the core claim fails. A finding of “no difference” is a real and publishable result.

8.2 Status of the evidence (stated plainly)

We have now run a first, deliberately minimal version of the experiment, designed so that it can fail. One hundred agents must track a drifting target they each sense noisily. The two arms differ in *one* thing only — the communication graph: a centralized *star* (a hub aggregates every report and broadcasts a single estimate, which followers adopt) versus a distributed *mesh* (each agent blends its own sensing with its neighbours’). No agent is hand-set to ignore information; the difference is purely topological. The same corruption attack hits both — a fraction of nodes keep reporting but emit a fixed wrong value (the degraded-but-trusted case of Layer 3) — under two adversaries: a *random* one that does not know the topology, and a *targeted* one that corrupts the most-connected node first. We sweep corruption from 0 to 50% over 200 random seeds per condition and measure coordination error (RMSE) during the attack.

The first thing to note is the control: with no corruption the *star wins* (RMSE 0.10 versus the mesh’s 0.53), because pooling many reports cancels noise. A model that favoured the distributed arm even here would be suspect; this one does not. As corruption appears, the ranking inverts, and how sharply depends on the adversary. Against a *targeted* adversary the star collapses to chance-level

error the moment any corruption is present and stays there, while the mesh degrades gracefully — roughly a threefold lower error at 10% corruption. Giving the star a robust (median) aggregator does not change this at all: when the corrupted node is the hub itself, no aggregation rule the hub runs can help, because the aggregator is the source of the lie. Against a *random* adversary the gap is real but milder, and the robust aggregator recovers some ground. The predicted trade-off (§10) thus appears as an earned crossover rather than an assertion.

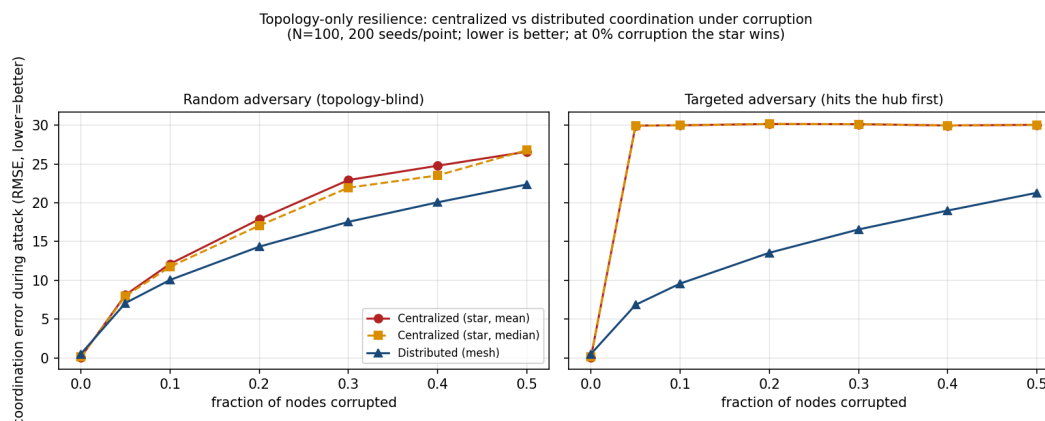


Figure 1. Coordination error during attack (RMSE; lower is better), $N=100$, 200 seeds/point. Left: a topology-blind (random) adversary. Right: a topology-aware (targeted) adversary. At 0% corruption the centralized star is best; under a targeted adversary it fails catastrophically and a robust median aggregator does not rescue it, while the distributed mesh degrades gracefully.

Condition	Star (mean)	Star (median)	Mesh
No corruption (control)	0.10	0.12	0.53
Targeted, 10% corrupted	29.98	29.98	9.59
Random, 30% corrupted	22.93	21.94	17.55

Scope and honest fences. This is a result *about this model*, not about crabs, armies, or firms. It tests the *topological* sub-claim — that a single aggregating node is a single point of failure under corruption — which is the mechanism behind the intent-based argument, not the full mission-command thesis: there is no intent-verification and no agency here, and the mesh is a stand-in for distributed decision-making. The targeted-star collapse is in part true by construction (corrupt the hub and the star must fail); the non-trivial findings are the clean-conditions crossover, the quantified random-versus-targeted gap, and the result that robust aggregation cannot save a compromised centre. One parameter set was used; a sweep is future work (§9). Earlier drafts circulated a “5–10×” figure and a 0.75→0.10 coherence drop; those specific numbers were never measured and are superseded by the RMSE results reported here. Code and data accompany the paper so the run can be reproduced and broken.

8.3 Why the architecture should help, layer by layer

- **Layer 1.** Intent is channel-agnostic: it survives in radio, runner, hand signal, or written note, so colonization of any one channel matters less.
- **Layer 2.** When subordinates are judged on achieving the intent, their private motivations tend to align with it rather than against it; divergence is easier to detect because it shows up as failure to advance the objective.

- **Layer 3.** With no detailed orders to corrupt and no single commanding node, there is less for a cascade to propagate through.
- **Layer 4.** A simple, stable intent is legible through noise; flooding degrades coordination but need not erase the objective.
- **Layer 5.** A saboteur's instruction can be checked locally against shared intent, so it is less able to command a cascade.

Each of these is an argued expectation, and each is something the experiment in §8.1 can support or undercut.

9. Open seams

The load-bearing weaknesses this draft does not resolve, collected honestly:

- **The simulation is a first pass.** §8.2 reports one model, one parameter set, testing one sub-claim (topology). The layer-compounding interaction, a clean corruption-versus-destruction comparison, a parameter sweep, and the full intent-based thesis all remain untested. A single run supports the modelled mechanism; it does not settle the broader argument.
- **The layers' compounding is asserted.** That vulnerability at one layer raises it at the next is intuitive but not derived; it needs a model in which the interaction is explicit and measurable.
- **One strong case, one contested one.** The wolfpack is documented; the contemporary case rests on contested reporting and may not survive better information. A theory resting heavily on a single clean case is exposed, and we say so.
- **"Corruption beats destruction" is a mechanism, not a measurement.** The asymmetry is plausible and has a clear logic, but its size — and whether it holds across regimes — is unknown until simulated or observed.
- **Intent-based command has costs the lens underweights.** It can be slower and less efficient in stable conditions, can fragment without a genuinely shared intent, and assumes subordinates competent and informed enough to interpret intent well. The trade-off in §10 is real and not always resolved in favour of distribution.

10. The trade-off, and where the lens applies

Nothing here says distributed always beats centralized. Centralized control is efficient and coherent in stable conditions: fewer decisions, clear lines, fast synchronization. Intent-based flexibility carries overhead — more local judgement, slower consensus, redundancy that looks wasteful until it is needed. The lens's actual claim is conditional: as the environment becomes more volatile and more adversarial — especially as adversaries shift from destroying nodes to corrupting them — the balance tips toward architectures that do not depend on perfect information. A crossover of exactly this kind — centralization better when the channel is clean, distribution better once it is corrupted — is what the topology experiment of §8.2 produces. This is the same instinct that runs through the resi-

lience literature in ecology and management: systems built to absorb disturbance and keep functioning, rather than to maximize efficiency in calm conditions, trade peacetime optimality for the ability to survive surprise [7, 8]. The practical question for any organization is which regime it is actually operating in, and whether it has mistaken a volatile one for a stable one.

Read this way, the cautious implications for various domains — militaries, firms, software systems, infrastructure — are variations on one sentence: where corruption of the channel is a credible threat, prefer designs that keep working when the channel is not trusted. We resist the stronger, tidier claims (that this explains the fall of empires, say) as illustrative analogies rather than results; civilizational collapse has many causes and is not adjudicated by a five-layer scheme.

11. Conclusion

The deepest version of the trap is not that a system is locked but that it does not know it is locked. The German command did not believe its cipher was read. A centralized force under stress may mistake the need to distribute for a need to centralize harder. Systems do not escape what they cannot see.

The escape we propose is correspondingly modest in its assumptions and, we think, demanding in practice: stop requiring the rung to be sound. Specify intent, distribute the authority to pursue it, keep more than one way to coordinate, and treat “we may already be compromised” as a working assumption rather than a worst case. That posture does not promise security. It promises something we think matters more under real conditions: function despite compromise. The first experiment reported here is a small down payment on that question; whether the posture delivers in full, at what cost, and by how much, is the empirical work this paper hands forward.

References

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- [7] Weick, K. E., & Sutcliffe, K. M. (2015). *Managing the Unexpected: Sustained Performance in a Complex World*. Wiley. (High-reliability organizations.)

[8] Walker, B., Holling, C. S., Carpenter, S. R., & Kinzig, A. (2004). Resilience, adaptability and transformability in social-ecological systems. *Ecology and Society* 9(2), 5.

Citation note: [1] and [2] are the author's companion deposits; cross-verify their DOIs and versions. [3], [4], [5], [7], and [8] are standard works cited from established literature; verify each against the original (years, volumes, pages) before external circulation. [6] anchors the Ultra material at the level of well-documented history rather than any quoted individual; confirm volume and pages, and the "Black May" loss figure, against the official history or other primary sources before citing specifics.

Referee Report

Self-review, doubling as the version tracker. Reviewer stance: skeptical but fair.

Summary judgment

A cross-domain synthesis lives or dies on whether it labels its own hierarchy — what is lens, what is claim, what would break it. This draft is structured to do that: frame-lock and the five layers are offered as an organizing lens (Parts 1–2), and credibility is staked on a single pulled-out claim with an experiment (§8). The chief risk in earlier drafts was that the paper's most memorable assertions — "corruption beats destruction," a "5–10×" resilience advantage — were unrun predictions reported as findings. That has been corrected: the specific old figures are withdrawn, and §8.2 now reports a first, fenced result from a topology-only run (with the model's clean-conditions advantage for centralization shown as a control). Recommendation: defensible as a speculative, openly-licensed synthesis with one earned model result and an honest account of what remains untested; the next steps are a parameter sweep and modelling the layer interaction.

Strongest and weakest

- **Strongest:** the False Rung as a unifying idea, and the Ultra case that anchors it. The concept — that unawareness of compromise converts vulnerability into self-harm — is sharp, documented, and travels across domains.
- **Weakest:** the empirical core now rests on a single first-pass model run (§8.2) and the paper leans on essentially one fully-documented historical case. The run tests only the topological sub-claim and uses one parameter set; the broader mission-command thesis, the layer-compounding, and a corruption-versus-destruction comparison are still untested. The contemporary case is correctly hedged and should perhaps be cut to a footnote if reporting shifts.
- **Honest exposure:** the layers' compounding is asserted, not modelled; "corruption beats destruction" is a mechanism awaiting measurement; the broader thesis's evidence does not yet exist. All are listed in §9.

For v5

- Sweep the §8.2 model over parameters (agent count, mesh connectivity, noise, drift, attack magnitude) to test how robust the crossover is.
- Model the layer-compounding interaction explicitly, so "compound" becomes a measured statement rather than a phrase.

- Verify [6] (Hinsley volumes) and the Black May loss figure against primary/official sources.
- Decide whether the contemporary case earns its place or belongs in a footnote.
- Consider a second fully-documented, non-biological case (industrial or doctrinal) to reduce reliance on the single wolfpack example.

Change log

v1-v2 (chat-assembled drafts) — framework and five layers established; cases drafted; simulator specified.

v3 — reformatted to the trilogy template (US Letter, navy/maroon heads, running footer). Epistemic pass: removed an invented quotation attributed to a real individual in the Ultra section and replaced it with sourced paraphrase; relabelled all simulator resilience numbers as *predictions*, *not findings* and added §8.2 (including a note on the question-begging failure mode of a naive design); changed “validate” to “illustrate” throughout; hedged the contemporary military case as contested open-source interpretation. Content pass: *removed the crab example and the orca contrast entirely* — the crab’s single-channel reading is complicated by documented fiddler-crab visual signalling and is better explained by kin-selection theory than by communication, and the orca “vocalization-evolved-first” story is an unfalsifiable just-so claim; both were illustrations the paper does not need. *Removed the fictional Prador coda and the Asher reference.* The paper now anchors on the documented Ultra/Enigma case, with Layer 1 illustrated by the wolfpack’s single radio substrate rather than by biology. Citations, the citation note, and keywords were updated accordingly.

v4 (this draft) — *ran the topology-only experiment and folded in the result.* Built a clean, non-rigged model (the two arms differ only in communication graph; the centralized arm is given its real clean-conditions advantage as a control), ran it over 200 seeds across a corruption sweep under random and targeted adversaries, and reported the measured crossover in §8.2 with a figure and table. Withdrew the old “5-10×” and 0.75→0.10 figures (never measured). Updated the abstract, framing note, §5.1, open seams, §10, the conclusion, and Appendix A from “unrun/prediction” to a fenced result, and scoped the result explicitly to the modelled topological sub-claim. Code and data (`topology_experiment.py`, `topology_results.json`) accompany the paper. Open for v5: parameter sweep; model the layer-compounding; corruption-versus-destruction comparison; reconsider the contemporary case; add a second documented case.

— end of v4 draft and referee report —

Appendix A: Simulator design (illustrative)

The accompanying simulator is an agent model intended to operationalize §8, not to stand as evidence on its own. It is released with the paper under CC BY 4.0 so that others can run, break, and extend it. A first, topology-only configuration has now been run and its result is reported in §8.2 (Figure 1); the design below is the fuller model the project is building toward, of which §8.2 exercises the topological core. Its design, in brief:

- **Agents.** Each carries a continuous pose, a small genotype (a combat-prowess parameter, a communication-fidelity parameter, a mutation rate), and a decision architecture flag (command-following vs. intent-based).
- **Environment.** A stochastic resource and threat field with diffusion and staleness, so that information has a shelf life.

- **Adversary.** A scripted attacker exposing the layers of §3: eavesdrop, jam, spoof, structural decapitation, feint, partial-compromise (mobility-kill), cascade, signal flood, and a trusted-saboteur mode.
- **Metrics.** A coherence measure combining spatial accuracy and temporal staleness; a trust/corruption index; and a recovery measure after attack.

The experiment of §8.1 compares the two architectures under matched attacks across many seeds. A fair comparison requires that the architectures differ structurally (topology) rather than by a parameter the attack directly removes; otherwise the result restates its premise. The defenses listed below are the design intent, themselves untested.

#	Attack	Mechanism	Layer	Design-intent defense
1	Eavesdrop	Read signals	1	Assume transparency; intent-based
2	Jam	Block signals	1	Redundant, dissimilar channels
3	Spoof	Inject false signals	1	Cross-validation against intent
4	Structural	Remove central node	2	Distributed authority
5	Feint	Mislead targets	2	Local verification
6	Partial-compromise	Mobility-kill / degrade	3	Isolation protocol
7	Cascade	Corrupt a trusted centre	3	No single commanding node
8	Signal flood	Overwhelm the channel	4	Legible standing intent
9	Saboteur	Trusted insider corrupts	2-5	Local check against shared intent

Code repository: <https://github.com/BareMetalHacker/frame-lock-hypothesis> · Contact: matt-schulz97@protonmail.com · ORCID: 0009-0000-0956-5743. The simulator is provided as-is and is, by design, something to be broken and improved rather than a finished instrument.